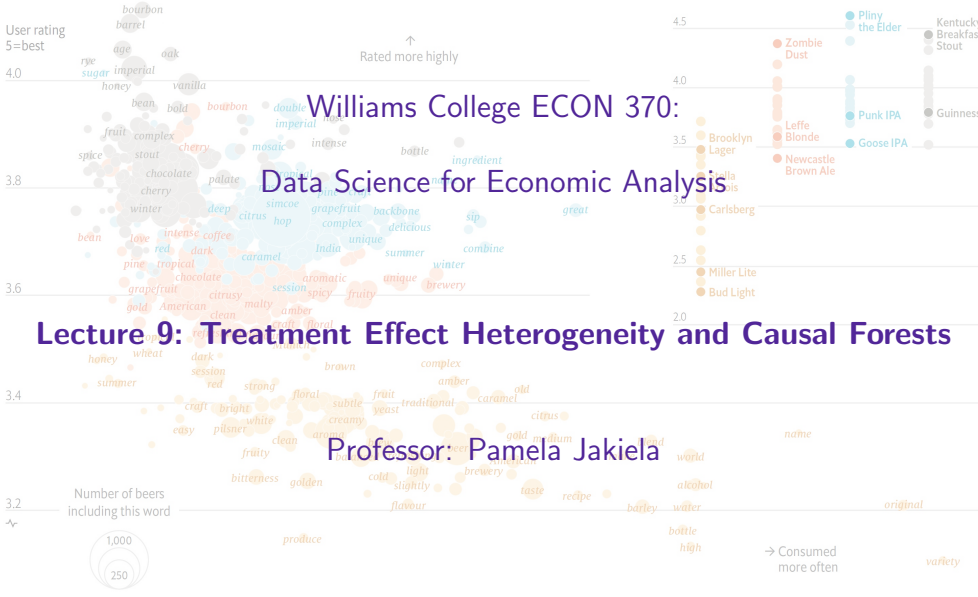


Williams College ECON 370:

Data Science for Economic Analysis

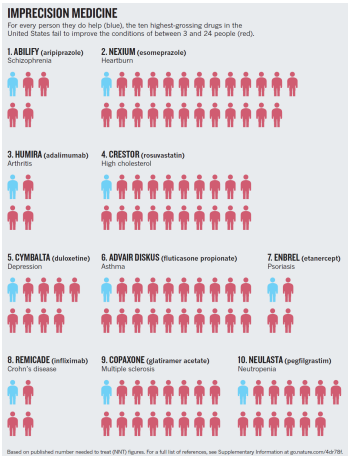
Lecture 9: Treatment Effect Heterogeneity and Causal Forests

Professor: Pamela Jakiela

Outline

- Regression approaches to treatment effect heterogeneity
- Causal trees and causal forests
- Generic machine learning to test for treatment effect heterogeneity

Motivation: The Promises and Pitfalls of Personalized Targeting



Source: Schork 2015

Motivation: The Promises and Pitfalls of Personalized Targeting

- Randomized controlled trials provide unbiased estimates of average treatment effects, but treatment effects may differ across subgroups defined by baseline characteristics
  - Medicine: both benefits of treatment and side effects
  - Firms: online targeting of both ads and interventions promoting user engagement
  - Microfinance loans: women vs. men, established entrepreneurs vs. ultra-poor borrowers
- Can we use baseline covariates (Xs) to identify high/low impact subgroups within RCTs?

Regression Estimates of the Average Treatment Effect (ATE)

Treatment effect on individual i: $\tau_i = Y_{1,i} - Y_{0,i} \Leftrightarrow Y_{1,i} = \tau_i + Y_{0,i}$

- We are often interested in the expected average treatment effect: $E[\tau] = E[Y_1] - E[Y_0]$

Estimate the OLS regression: $Y = \alpha + \beta D$

- $\hat{\alpha} = \bar{Y}_C$, the average outcome in the control group
- $\hat{\beta} = \bar{Y}_T - \bar{Y}_C$, the difference in means between the treatment and control groups

The OLS coefficient $\hat{\beta}$ provides an unbiased estimator of the average treatment effect

$$E[\hat{\beta}] = E[\bar{Y}_T] - E[\bar{Y}_C] = \ldots = E[Y_1|D = 1] - E[Y_0|D = 0] = \ldots = E[\tau]$$

because $E[Y_0|D = 1] = E[Y_0|D = 0] = E[Y_0]$ when treatment is randomly assigned

Conditional Average Treatment Effects (CATE)

$E[\bar{Y}_T|X = x] - E[\bar{Y}_C|X = x]$ provides an estimate of $E[\tau|X = x]$, the ATE when $X = x$

- Under random assignment, $E[X|D = 1] = E[X|D = 0] = E[X]$

Interacted specification: $Y = \alpha + \beta D + \theta X + \delta (D \times X)$

- X is a dummy variable capturing a baseline characteristic (e.g. sex or poverty status)
- $\hat{\alpha} =$
- $\hat{\beta} =$
- $\hat{\theta} =$
- $\hat{\delta} =$

$\Rightarrow$ Estimated treatment effect on those with $X = 1$:

Treatment Effect Heterogeneity: EMERGE Example



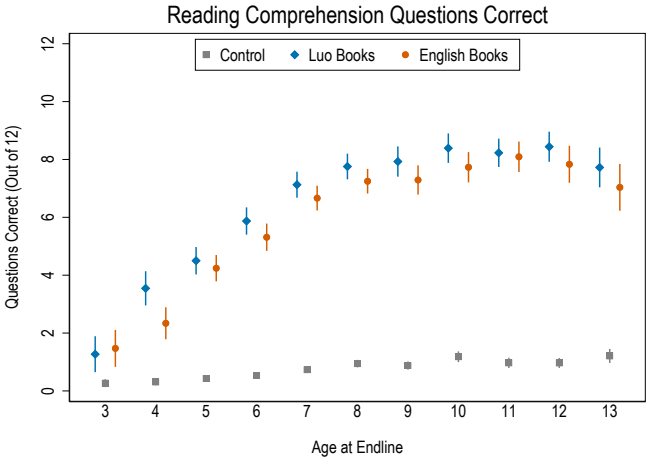
What 3 things did Caro forget at home today?



What do you think Meg and Ben are looking at?

We developed an objective measure of children's use of storybooks:
12 comprehension questions that children would be unlikely to guess

Treatment Effect Heterogeneity: EMERGE Example



Treatment Effect Heterogeneity: EMERGE Example

Outcome:	**Storybook Comprehension**		
	(1)	(2)	(3)
Treatment	5.519	4.682	1.481
	(0.131)	(0.168)	(0.417)
	[0.000]	[0.000]	[0.001]
Older (age $\geq$ 59 months)		0.378	
		(0.047)	
		[0.000]	
Treatment $\times$ Older		1.640	
		(0.163)	
		[0.000]	
Age in months			0.015
			(0.002)
			[0.000]
Treatment $\times$ Age in months			0.068
			(0.006)
			[0.000]
Observations	2,200	2,200	2,200

Standard errors, clustered at the school level, in parentheses; p-values in square brackets. All regressions include fixed effects for randomization strata. **Older** is an indicator for children at or above the median age of 59 months. Age in months is not centered, so the treatment coefficient in column (3) is the effect at age zero.

Treatment Effect Heterogeneity with Sparse Data

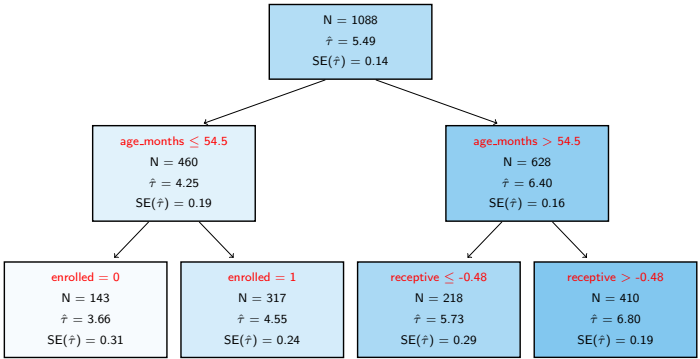
Thought experiment: you test for treatment effect heterogeneity using 100 baseline variables

- How many variables do you expect to be statistically significant (p-values < 0.05)?

The "old-fashioned" approach to testing for treatment effect heterogeneity works very well (and has good statistical power) if you can identify key dimensions of heterogeneity in advance

- We often want a more flexible approach that allows us to search for heterogeneity
- But we also want to avoid **p-hacking** (don't test 100 variables and report 2 of them!)
- Machine learning allows us to identify Xs that predict large differences in the mean of Y between the treatment and control groups (the treatment effect heterogeneity problem)

From Regression Trees to Causal Forests



A **causal forest** is a machine learning algorithm for identifying treatment effect heterogeneity (Athey and Imbens 2016, Davis and Heller 2017, Wager and Athey 2018)

A Causal Tree

When building a regression tree, we partition the sample into test data and training data

- We use the training data to build the tree through recursive binary splitting
  - Splits are chosen to minimize the MSE given the leaf-specific predictions of Y (call them $\hat{\mu}_\ell$)
  - This is mathematically equivalent to maximizing the variance of $\hat{\mu}_\ell$
- We use the test data to calculate test MSE: $\frac{1}{N_{\text{test}}} \sum_{i \in \text{test}} (Y_i - \hat{\mu}_{\ell(i)})^2$

In building a **causal tree**, we seek to minimize the MSE of the predicted treatment effects, $\hat{\tau}_i$

- We don't know τ_i for any i (because of the fundamental problem of causal inference), so we cannot calculate the MSE of potential splits to determine which one minimizes MSE
- However, minimizing MSE is equivalent to maximizing the variance of the $\hat{\tau}_\ell$ predictions

An Honest Tree

When building a (standard) regression tree, we split the sample into test data and training data

- We use the training data to build the tree and make predictions
- We use the test data to calculate test MSE

The same training data is used to build the tree and calculate the leaf-specific $\hat{\tau}_\ell$ predictions

$\Rightarrow$ Predictions of τ are biased, not independent of errors in the training data

In an **honest** regression tree, we split the sample into training, estimation, and test samples

$\Rightarrow$ Predictions are independent of observation-specific errors that influence tree splitting

With an honest causal tree (or forest), we can calculate confidence intervals around our predictions of conditional average treatment effects using conventional inference approaches

Causal Forests in R and Python

- Causal forests are implemented as double ML (relevant when treatment is not random)
  - Use X variables to predict outcome Y and treatment W
  - Train the causal forest on residualized $\tilde{Y} = Y - \hat{Y}$ and $\tilde{W} = W - \hat{W}$
    - In R (`grf`), we can set the predicted values of Y and W to 0 or the mean (in an RCT)
    - In Python (`econml`), we use sklearn's `DummyRegressor` and `DummyClassifier`
- Tuning parameters are similar to regression trees and random forests
  - Number of trees, minimum leaf size, depth (Python only)
  - Proportion treated within a leaf is also relevant with trees

An EMERGE Causal Forest in R: Syntax Example

```
install.packages("grf")
library(grf)

cf <- causal_forest(X, # X variables
                    Y, # outcome
                    W, # treatment variable
                    Y.hat = Y_hat, # model to use to predict Y
                    W.hat = W_hat, # model to use to predict treatment
                    clusters = village,
                    tune.parameters = "all",
                    num.trees = 200,
                    seed = 8675309
                    )
```

Additional information on options and tuning parameters is available at:
https://grf-labs.github.io/grf/reference/causal_forest.html

Interpreting Causal Forests

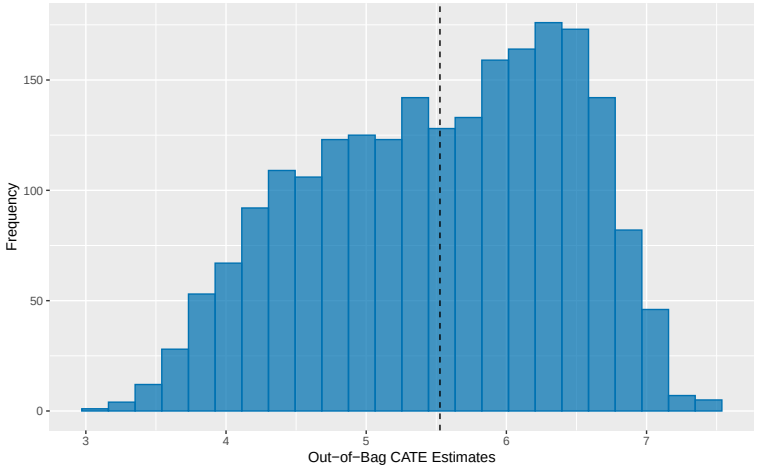
Causal forests (can) tell us two things:

- Is there (meaningful) treatment effect heterogeneity **that is explained by observables**?
  - Machine learning is noisy in small samples!
  - Algorithm looks for heterogeneity **that is predicted by baseline covariates**
- Which baseline covariates predict the observed treatment effect heterogeneity?

Answerable questions and relevant outputs from causal forests:

- What are the observation-level out-of-bag (OOB) predicted CATEs?
- Does the variation in out-of-bag CATEs explain variation in Y (`test_calibration` in R)?
- Are estimated treatment effects higher for observations w/ high OOB CATEs?
- How often a covariate was used to split a tree (variable importance)?

Distribution of Out-of-Bag CATE Estimates: Example

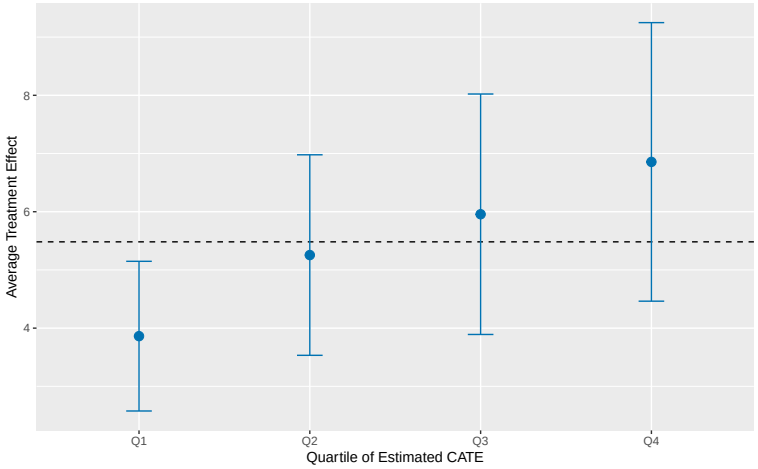


Is There (Statistically Significant) Treatment Effect Heterogeneity?

Partition sample into quantiles based on out-of-bag CATE estimates

- Is ATE in highest quantile (significantly) different from ATE in lowest quantile?
- Caveat: sorting by estimated CATEs and then calculating standard errors is not "honest"

Group Average Treatment Effects: Example



Is There (Statistically Significant) Treatment Effect Heterogeneity?

Partition sample into quantiles based on out-of-bag CATE estimates

- Is ATE in highest quantile (significantly) different from ATE in lowest quantile?
- Caveat: sorting by estimated CATEs and then calculating standard errors is not "honest"

Alternative approach: do CATE estimates explain variation in Y?

- Generate two variables: $\bar{\tau}$ (ATE) and $\hat{\tau}_i$ (individual predicted CATE estimate)
- Regress Y on $\bar{\tau} D_i$ and $(\hat{\tau}_i - \bar{\tau})\, D_i$
- Coefficient on $(\hat{\tau}_i - \bar{\tau})\, D_i$ tests the hypothesis that variation in individual-level predicted CATEs explains variation in the outcome in the treatment group (relative to control)

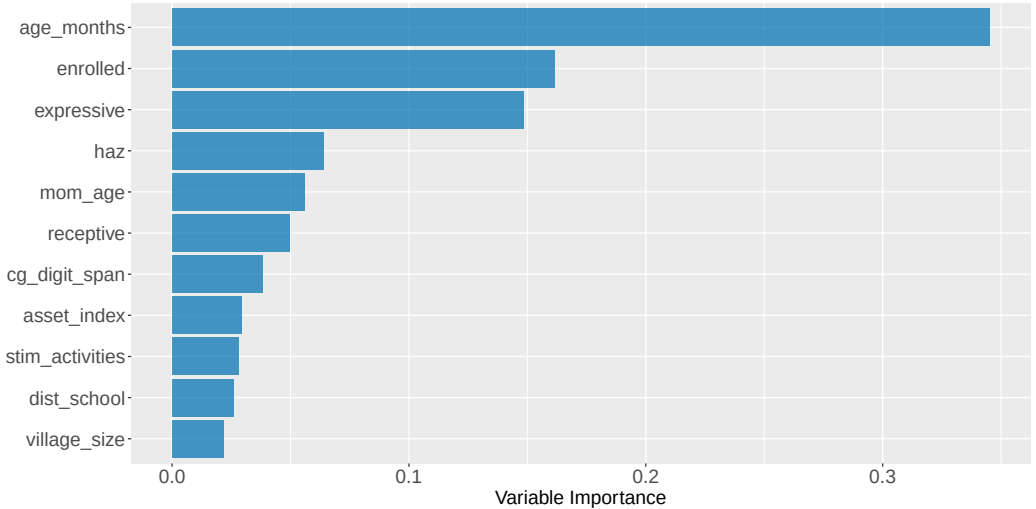
Testing for Treatment Effect Heterogeneity: Example

```
cf_test <- test_calibration(cf)
cf_test

Best linear fit using forest predictions (on held-out data)
as well as the mean forest prediction as regressors, along
with one-sided heteroskedasticity-robust (HC3) SEs:

                                Estimate  Std. Error  t value    Pr(>t)
mean.forest.prediction           0.99212     0.16180   6.1317  5.138e-10  ***
differential.forest.prediction   1.35346     0.32251   4.1967  1.408e-05  ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Variable Importance in Causal Forests: Example



Causal Forests: Summary

A causal forest is a machine learning algorithm for detecting treatment effect heterogeneity

- A (causal) forest is made up of many (causal) trees
- Build a causal tree by choosing splits to maximize the variance of $\hat{\tau}_\ell$ predictions

A causal forest allows us to test for treatment effect heterogeneity predicted by covariates

- What does a histogram of the out-of-bag CATE estimates look like?
- Do the predicted CATEs explain variation in Y in the treatment group relative to control?
- Which variables are important predictors of treatment effect heterogeneity?

Generic Machine Learning Approach

Test for heterogeneous treatment effects following Chernozhukov et al. (2025)

- Propose a **generic machine learning** approach that allows for inference about **key features** of treatment effect heterogeneity, can be used with any machine learner(s)
- Use machine learning to predict potential outcomes with and without treatment, and hence implicitly predict the treatment effect as a function of baseline characteristics
- Use predictions to:
  1. Test for (predictable) treatment effect heterogeneity
  2. Test for differences in impact between most-impacted and least-impacted groups
  3. Generate individual-level proxies for estimated **conditional average treatment effects**

Generic Machine Learning: Setup

Approach requires the following:

- Outcome: Y
- Baseline characteristics: Z
- Randomly-assigned treatment: D

Split the sample, use any ML approach to estimate (auxiliary data) and predict (main data):

- $B(Z)$: proxy for the **baseline conditional average**: $b_0(Z) = E[Y_0|Z]$
- $S(Z)$: proxy for the **conditional average treatment effect**: $s_0(Z) = E[Y_1|Z] - E[Y_0|Z]$

$B(Z)$ and $S(Z)$ can be estimated using any supervised learning approach, but we restrict attention to elastic net and random forest (other learners do not increase statistical power)

Testing for Treatment Effect Heterogeneity

Is there heterogeneity in $s_0(Z)$ that can be predicted by $S(Z)$?

- Estimate the regression: $Y = \beta_0 + \beta_1 B(Z) + \beta_2 D + \beta_3 D\,[S(Z) - E[S(Z)]]$
  - $\hat{\beta}_2$ provides an (alternate) estimate of the average treatment effect
  - $\hat{\beta}_3$ provides a test of the joint hypothesis that there is treatment effect heterogeneity **and** that treatment effect heterogeneity can be explained by the baseline characteristics

Testing for Differences in Treatment Effects Across Groups

Use $S(Z)$ to partition the sample into quartiles based on predicted treatment effects

- For $k = 1, 2, 3, 4$, let G_k be an indicator equal to one for observations in quartile k
- Define the **sorted group average treatment effect** in quartile k: $\gamma_k = E[s_0(Z)|G_k = 1]$

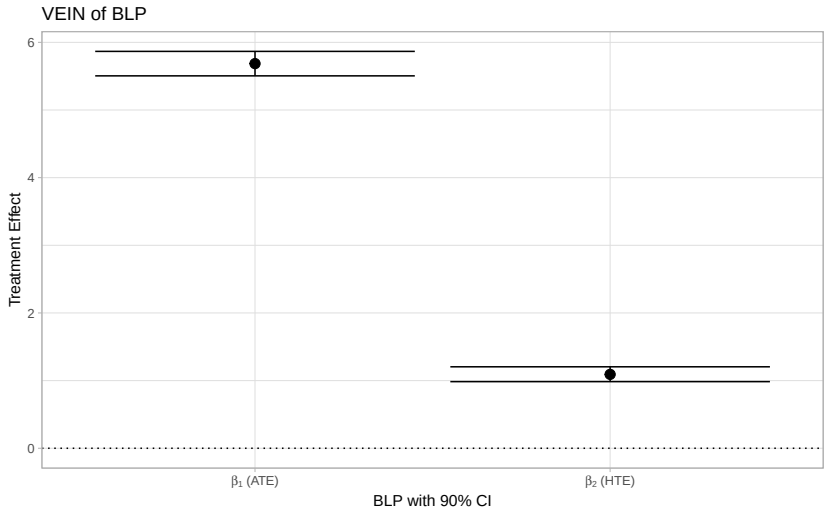
Estimate the regression equation:

$$Y = \theta_0 + \theta_1 B(Z) + \sum_{k=1}^{4} \alpha_k DG_k$$

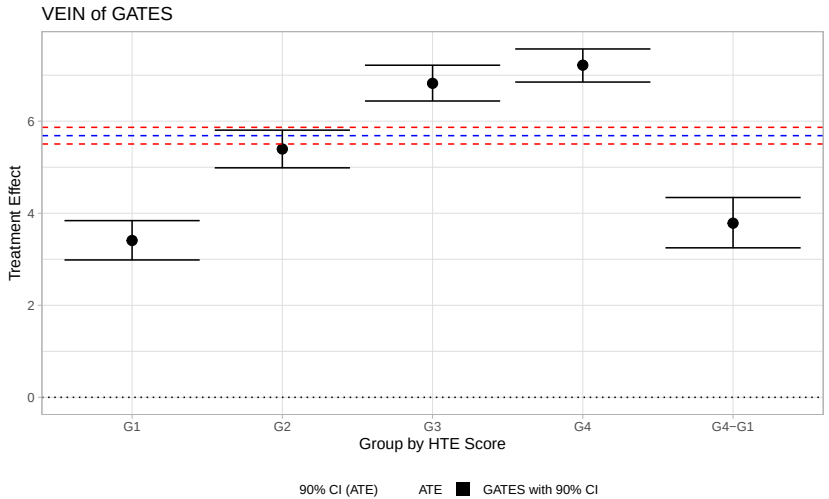
- For $k = 1, 2, 3, 4$, $\alpha_k = \gamma_k$, providing a test of the null hypothesis of no effect on Group k
- Testing the hypothesis that $\alpha_1 = \alpha_4$ is another test of treatment effect heterogeneity

Use a similar approach to test for differences in baseline Z variables across the same groups

Testing for Treatment Effect Heterogeneity Using GML



Sorted Group Average Treatment Effects



Characteristics of Most-Impacted and Least-Impacted Groups

